Blood Glucose Prediction Using Bayesian Neural Network

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Project Statement :

Accurate blood glucose prediction is essential for effective management of Type 1 Diabetes (T1D). However, most existing approaches focus primarily on improving predictive accuracy while overlooking the importance of quantifying uncertainty, which is critical for safe and informed clinical decision-making.

In practice, relying solely on a single predicted value can be risky, especially during rapid glucose fluctuations where the likelihood of adverse events such as hypoglycemia increases. Without an understanding of prediction confidence, such models provide limited clinical value.

This project addresses this gap by developing an uncertainty-aware blood glucose prediction framework using a Bayesian neural network. The proposed approach not only aims to improve prediction accuracy but also provides calibrated uncertainty estimates, enabling more reliable and risk-aware decision-making.

Project Objectives :

The primary objective of this project is to design and implement an uncertainty-aware blood glucose prediction system using Continuous Glucose Monitoring (CGM) data. The system aims to improve both predictive performance and reliability by integrating statistical inference techniques with Bayesian deep learning.

Specifically, the project seeks to:

- Develop a time-series forecasting model to predict blood glucose levels 30 minutes ahead using historical CGM data
- Incorporate Bayesian learning to estimate both prediction mean and uncertainty (aleatoric and epistemic)
- Quantify uncertainty through probabilistic outputs such as variance and credible intervals for safer clinical interpretation
- To apply and integrate classical and Bayesian statistical inference methods including MLE, MME, Fisher Information, Cramér–Rao bounds, conjugate priors, MAP estimation, credible intervals, and UMVUE (via the Lehmann–Scheffe theorem) to analyze data distributions, estimate parameters, and ensure efficient and theoretically sound modeling
- Build a deep learning architecture (BiGRU with attention and Bayesian layers) to capture temporal dependencies and uncertainty
- Evaluate model performance using both standard metrics (RMSE, MAE, R^2) and uncertainty-aware analysis
- Provide a foundation for risk-aware decision support in diabetes management by combining prediction accuracy with confidence estimation
- Demonstrated the model through an interactive web application that accepts real glucose history and returns predictions with full uncertainty decomposition.

Dataset- OhioT1DM

It contains eight weeks of continuous glucose monitoring (CGM) data recorded at 5-minute intervals for six Type 1 Diabetes patients, comprising 69,147 training sequences and 15,862 test sequences for glucose prediction tasks.

https://www.kaggle.com/datasets/ryanmouton/ohiot1dm/data

Patients	6 (T1D)
Total training observations	69,147 sequences
Total test observations	15,862 sequences
Sampling interval	5 minutes (CGM)
Input window (SEQ_LEN)	24 readings = 2 hrs
Glucose range (training)	~40 – 400 mg/dL
MLE mean (μ)	160.13 mg/dL

Dataset split : OhioT1DM split XML files labelled training form the training set and those labelled testing form the test set.

- Training set: 80% of training XML data — used for gradient updates
- Validation set: 20% of training XML data — used for early stopping
- Test set: Official OhioT1DM test XMLs — used only for final evaluation

Project Pipeline :

- Data Acquisition & Parsing - Import raw OhioT1DM XML files and parse them into structured tabular formats, isolating individual patient timelines to prevent data leakage.

- Preprocessing & Cleaning - Handle missing values via linear interpolation and filter out invalid blood glucose readings.

- Statistical Feature Engineering - Compute rolling statistical moments (mean, standard deviation, min, max) and temporal differences (lags) strictly
  within each patient's data boundaries.

- Sequence Generation - Transform the 1D time-series data into 3D tensors (24-step historical windows) and apply standard scaling fitted exclusively on the training set.

- Bayesian Model Construction - Build the neural network architecture featuring the 3-layer Bidirectional GRU (BiGRU), the Attention Mechanism, and the custom Bayesian Linear layers.

- Probabilistic Training - Optimize the model using a custom NLL (Negative Log-Likelihood) + KL Divergence loss function, employing early stopping and learning rate scheduling to prevent overfitting.

- Inference & Uncertainty Quantification - Generate predictions on unseen data, calculating both Aleatoric (data) and Epistemic (model) uncertainty using Monte Carlo Dropout to form 95% Credible Intervals.

- Clinical Risk Evaluation - Evaluate the model's performance using asymmetric LINEX loss to account for the differing clinical risks of hypoglycemia versus hyperglycemia.

- Dashboard Integration (GlucoSense) - Deploy the model to a backend service and connect it to the frontend UI, allowing users to input 120-minute histories and visualize the probabilistic 30-minute forecast.

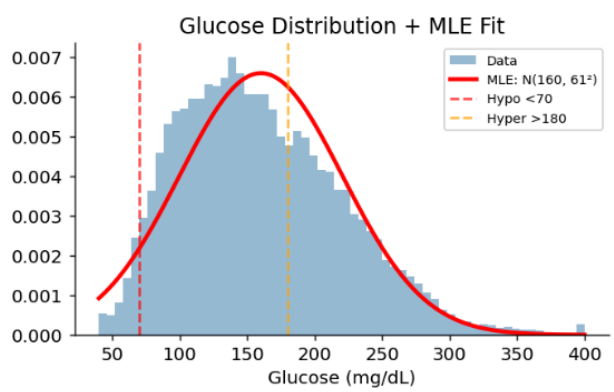
Statistical Foundations & Exploratory Data Analysis (EDA) :

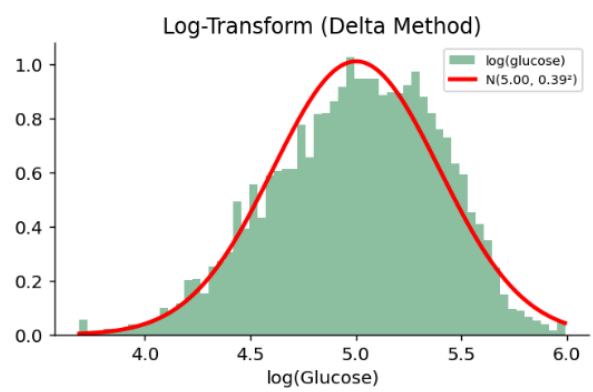
Before applying a Bayesian deep learning model, it is essential to understand the statistical properties of the glucose data. This step establishes baseline distributions, validates assumptions, and provides theoretical limits for prediction and uncertainty estimation.
Parameter Estimation (MLE & MME)
MLE and MME consistently model the glucose data as a Normal distribution. These estimates establish the global mean and variance used by the StandardScaler to center the 120-minute input sequences at zero, ensuring efficient BiGRU training and preventing exploding gradients.

```
============================================================
  MLE & MME ESTIMATION
============================================================
  n (training observations)  = 69255
  MLE  mu                    = 160.1300 mg/dL
  MLE  sigma                 = 60.5376 mg/dL
  MME  mu  (1st moment)      = 160.1300 mg/dL
  MME  sigma (2nd moment)    = 60.5376 mg/dL
```

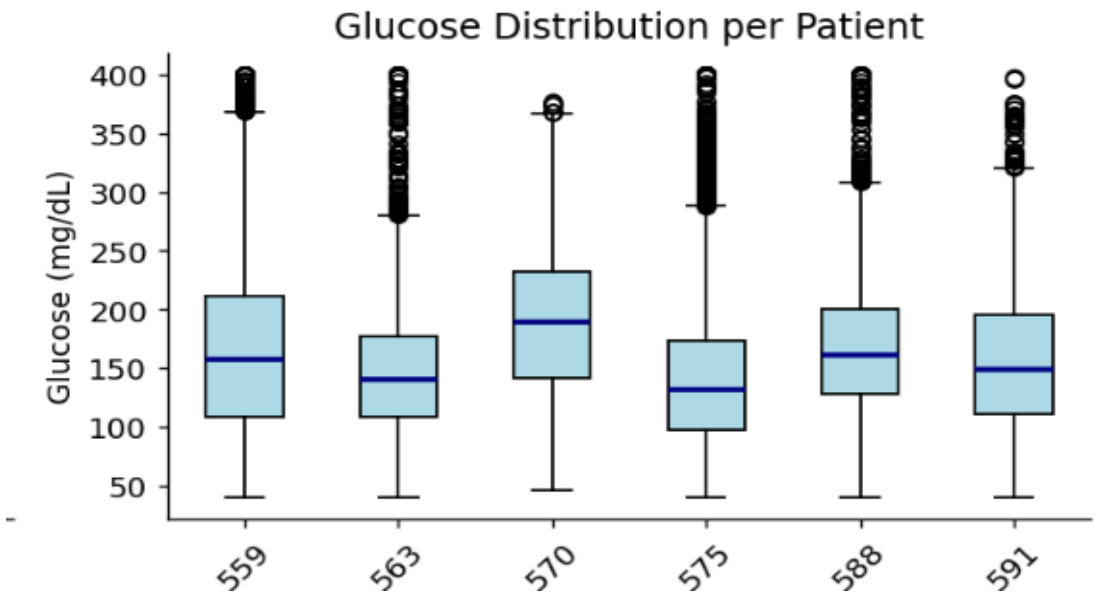
Log-Transform (Delta Method) : This plot shows the glucose data after a log-transformation to achieve a more symmetric, "bell-shaped" distribution. It justifies using Negative Log-Likelihood (NLL) as the model's loss function, as the data becomes mathematically easier to model in this transformed space.

Glucose Distribution + MLE Fit: This graph uses Maximum Likelihood Estimation (MLE) to fit a Normal distribution curve to the raw glucose data.



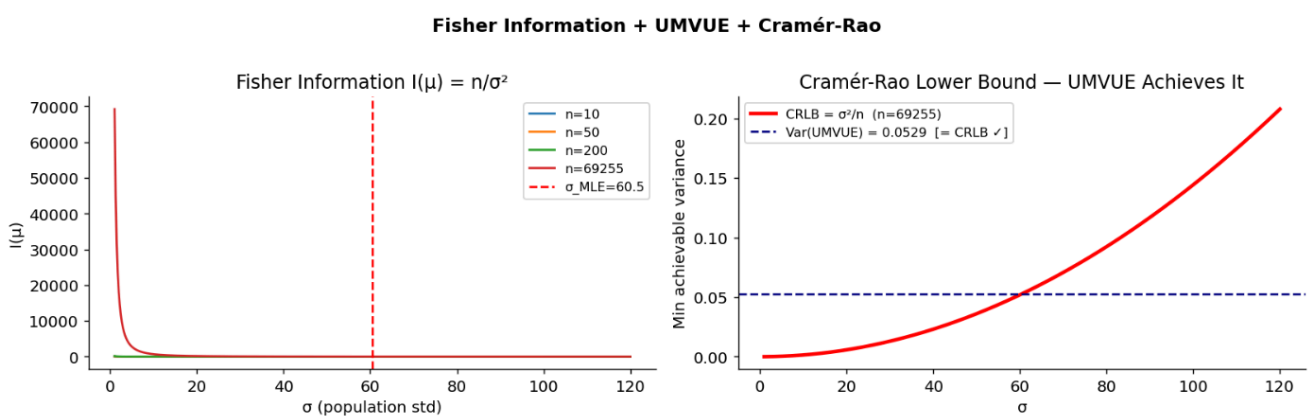


Glucose Distribution per Patient:



Fisher Information and Cramér–Rao Lower Bound

```
  FISHER INFORMATION
==========================================
  I(mu)          = n/sigma^2    = 18.897350
  I(sigma^2)     = n/(2sig^4)   = 0.00257822
  CRLB for mu    = 1/I(mu)      = 0.052917
  CRLB for s^2 = 1/I(sig^2)   = 387.8639
  FIM:
[[1.88973496e+01 0.00000000e+00]
 [0.00000000e+00 2.57822411e-03]]
```



```
  T (sufficient statistic) = Σxi  = 11089804.00
  UMVUE = T/n = x̄            = 160.1300 mg/dL
  Var(UMVUE) = sigma^2/n     = 0.052917   [= CRLB ✓]
  SE(UMVUE)  = sigma/sqrt(n) = 0.230038

STEP 3 — Cramér-Rao Lower Bound verification:
  CRLB = 1/I(mu) = sigma^2/n = 0.052917
  Var(UMVUE)                 = 0.052917
  UMVUE achieves CRLB: True
```

Fisher Information measures how much information the data provides about the parameter (mu). For this dataset, (I(mu) =18.897), indicating good estimation strength.

The Cramér–Rao Lower Bound (CRLB) gives the minimum possible variance of any unbiased estimator. Here, (CRLB= 0.0529).

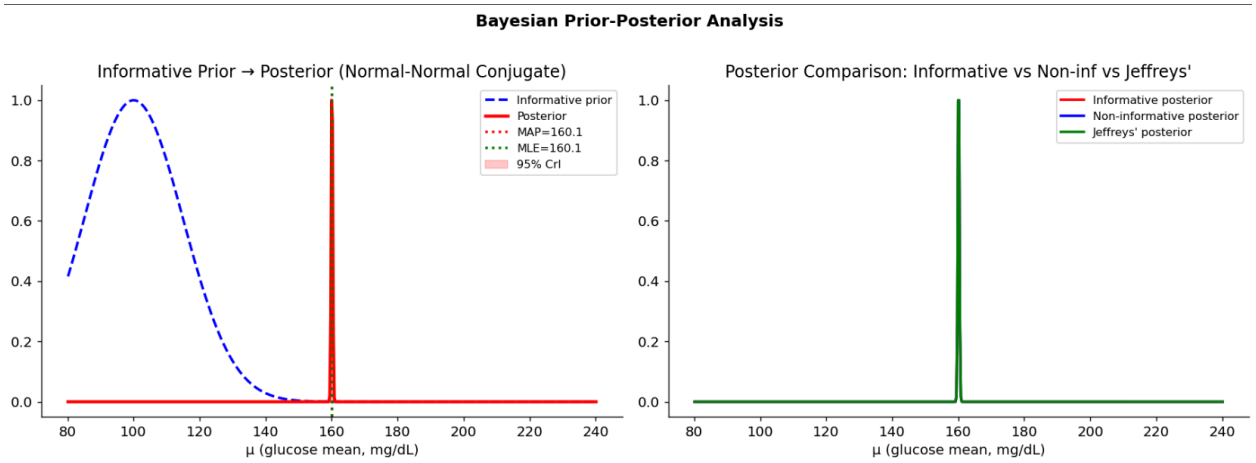
Since the variance of the estimator (UMVUE / sample mean) equals the CRLB, the estimator is efficient and achieves the optimal performance.

The graph shows that as noise (sigma) increases, Fisher Information decreases and CRLB increases, meaning estimation becomes less precise.

Overall, the estimator operates at the theoretical limit of accuracy for this data.

Bayesian Prior & Posterior Analysis

Proves that incorporating informative clinical priors stabilizes estimates against noisy data. This mathematically justifies the network's Bayesian architecture (specifically the KL Divergence loss), ensuring predictions remain safely constrained by clinical reality rather than overfitting to erratic sensor noise.




```
[1] Informative Prior: mu0=100.0, tau0=15.0
    Posterior mean (= MAP) = 160.1159 mg/dL
    Posterior median       = 160.1159 mg/dL
    Posterior std tau_n    = 0.230011
    95% Credible Interval  = (159.6651, 160.5667)

[2] Non-Informative Prior: mu0=0.0, tau0=10000.0
    Posterior mean (= MAP) = 160.1300 mg/dL  (≈ MLE = 160.1300)
    95% Credible Interval  = (159.6791, 160.5809)

[3] Jeffreys' Prior: p(mu) ∝ sqrt(I(mu)) = const [improper]
    Posterior mean         = 160.1300 mg/dL  (= MLE)
    Posterior std          = 0.230038
    95% Credible Interval  = (159.6791, 160.5809)

[4] Frequentist 95% CI
    t_crit = 1.959998
    CI = (159.6791, 160.5809)
```

Uncertainty & Interval Estimation:

Uncertainty in glucose predictions is represented using probabilistic intervals derived from the model. These intervals provide a range of possible outcomes, allowing users to assess risk instead of relying on a single predicted value.

Two approaches are compared:

- Frequentist Confidence Interval (CI)
- Bayesian Credible Interval (CrI)

```
=== CI vs CrI Interpretation ===
  Freq CI    (159.68, 160.58): If repeated, 95% of intervals contain true mu
  Bayes CrI  (159.67, 160.57): P(mu in interval | data) = 0.95
```

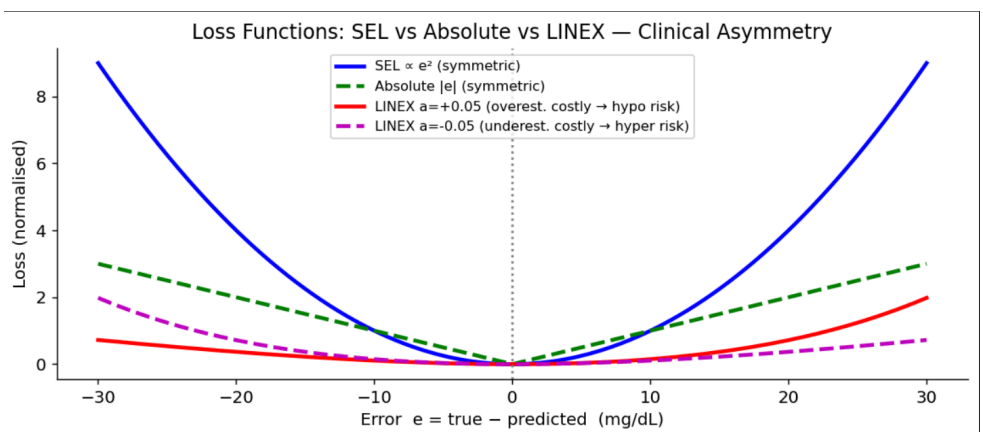
Asymmetric Risk Evaluation (LINEX vs. Symmetric Loss):

MSE and MAE are symmetric loss functions that treat overestimation and underestimation equally. In contrast, LINEX loss is asymmetric and penalizes errors differently based on their direction.

The behavior of LINEX depends on parameter (a):

- (a > 0): overestimation penalized more
- (a < 0): underestimation penalized more

In clinical glucose prediction, underestimating glucose (missing hypoglycemia) is more dangerous. Hence, using LINEX with (a < 0) helps the model prioritize patient safety by penalizing such errors more heavily.



DataPreprocessing: Preprocessing phase involves Feature Scaling and Sequence Reconstruction, Linear Interpolation.

- **Handling Missing Values:** We used Linear Interpolation (drawing a straight mathematical line between the last known point and the next known point). We dropped any rows containing NaN (Not a Number).During the parsing stage, any raw glucose reading of 10 mg/dL or below was considered a physical hardware error and was completely dropped before entering the dataset.
- **Data Scaling (Standardization)**: Used StandardScaler to ensure all 14 features have a mean of 0 and a variance of 1.
- **Sliding Window Sequence Building**: The function converts the flat data into sliding windows of 24 time steps. Sliding Window algorithm restructure the data into 3D tensors (Batch, Sequence, Features). Each sequence consists of 24 historical steps.

Feature Engineering:

It takes a single number (current glucose) and expands it into 14 different dimensions. The raw Continuous Glucose Monitor (CGM) data was transformed into a multi-dimensional feature space to capture the underlying metabolic state of each patient. To provide the BiGRU model with sufficient temporal context, Historical Memory was established through a series of lag features , allowing the architecture to analyze glucose trends over the preceding hour. Trend Analysis was incorporated by calculating first-order differences (velocity), which explicitly define the rate of change and direction of glucose movement. Furthermore, Stability Metrics were derived using

rolling windows to compute the standard deviation, maximum, and minimum values over 30 and 60-minute intervals. These rolling statistics provide a measure of glycaemic volatility, enabling the model to distinguish between steady-state glucose levels and high-risk, rapid fluctuations.

Bayesian Bidirectional GRU :

A Bayesian Neural Network (BNN) treats model parameters (weights and biases) as probability distributions instead of fixed values. Instead of learning a single weight, the model learns a distribution (mean and variance) for each parameter.

Starting with a prior distribution (e.g., Normal centered at zero), the model updates its beliefs using data to obtain the posterior $P(w \mid D)$.

During inference, weights are sampled from these distributions, producing slightly different outputs each time. This allows the model to provide not just predictions but also uncertainty estimates (credible intervals), indicating the range in which the true value is likely to lie.

ModelArchitecture:

3-Layer Bidirectional GRU (BiGRU)

- Sequential Processing: The model uses three layers of Gated Recurrent Units (GRU) to process 24 historical glucose readings (2-hour window).

- Bidirectional Flow: By being "bidirectional," the model reads the glucose sequence both forwards and backwards, allowing it to capture complex trends and slopes.
- Hidden Dimension: Each layer has a hidden size of 50 units, providing enough capacity to learn patient-specific metabolic patterns.

2. Attention Mechanism

- Feature Weighting: After the BiGRU layers, an Attention block assigns different "importance scores" to each of the 24 time steps.
- Context Vector: It identifies which specific readings (e.g., a sudden drop 15 minutes ago) are most critical for predicting the glucose level 30 minutes into the future.

3. Bayesian Output Layer (BayesianLinear)

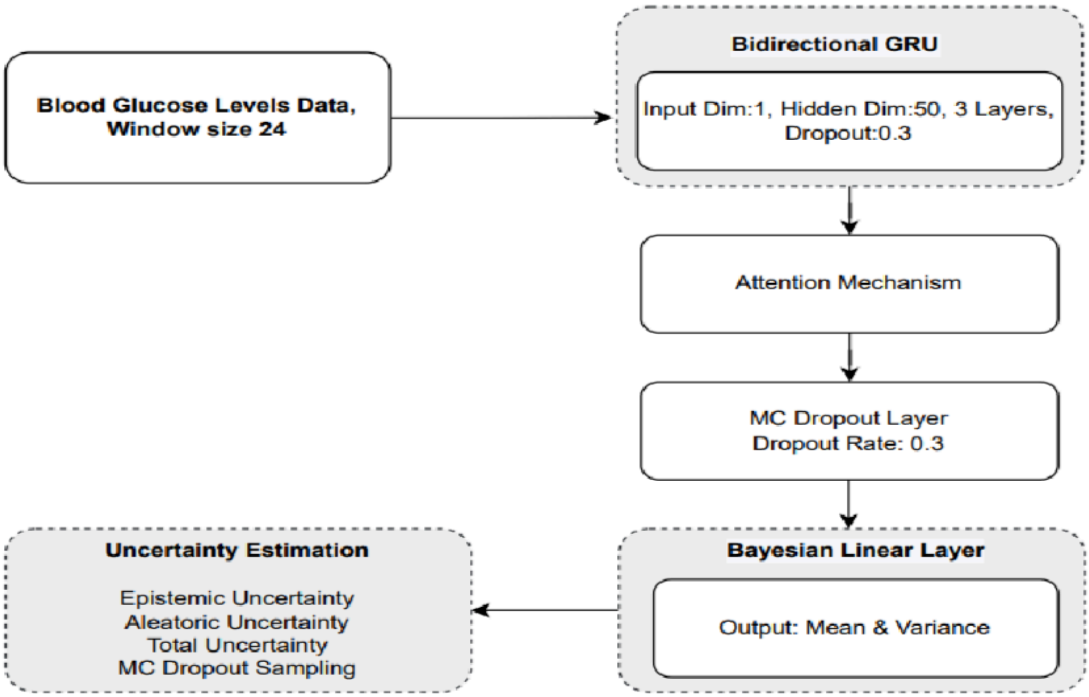
- Probabilistic Heads: Unlike a standard model that outputs one number, your final layer is a BayesianLinear layer that produces two distinct outputs: a mean and a log-variance.
- Weights as Distributions: This layer treats its weights as probability distributions, which is what allows you to perform Monte Carlo Dropout to measure uncertainty.
- Dual Uncertainty: This specific design is what enables the model to separate Epistemic uncertainty (model confusion) from Aleatoric uncertainty (sensor noise).

4. Training Logic

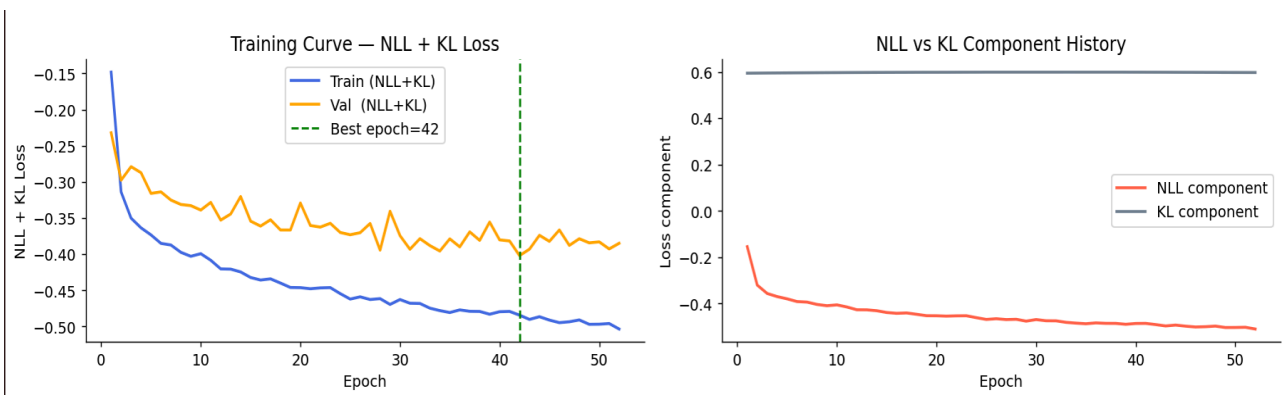
- NLL + KL Loss: Then trained this architecture using a custom loss function that combines Negative Log-Likelihood (NLL) to ensure accuracy and a KL Divergence term to keep the Bayesian weight distributions mathematically sound.
- Horizon: The entire architecture is tuned for a 30-minute prediction horizon, providing a clinically useful early-warning system for the patient.

Training: Train: 55202 | Validation: 13801 | Test: 15718

Metric	Value	Benchmark
Optimizer	Adam	lr = 0.0001
Loss function	NLL + KL	KL weight = 0.01
Max epochs	50	50
Early stopping	Patience = 15	Best epoch: 42



Training Curve:

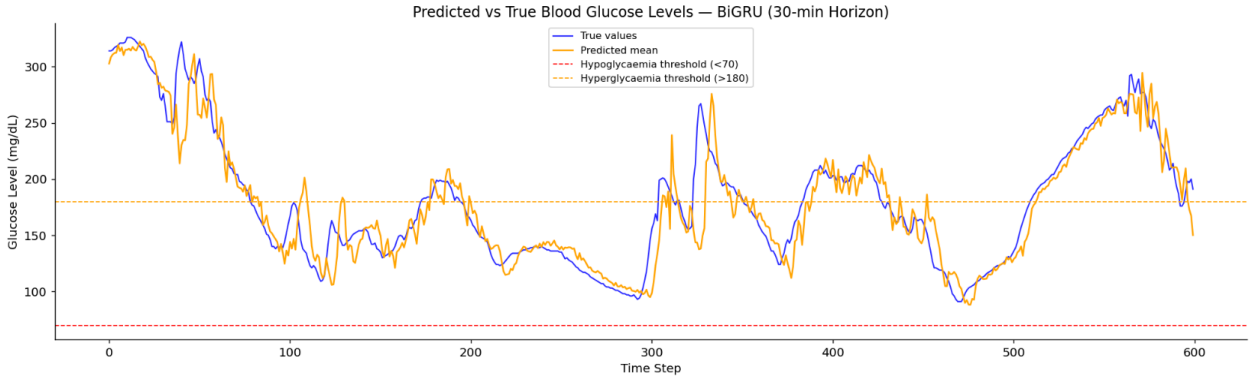


- The training loss decreases steadily across epochs, indicating stable learning. Early stopping halts training at epoch 42, where validation loss is minimized, helping prevent overfitting.
- The KL divergence component remains relatively stable throughout training, showing consistent regularization of the Bayesian weight distributions toward the prior $N(0,1)$, without dominating the loss.

Monte Carlo Dropout Inference and Uncertainty Quantification :

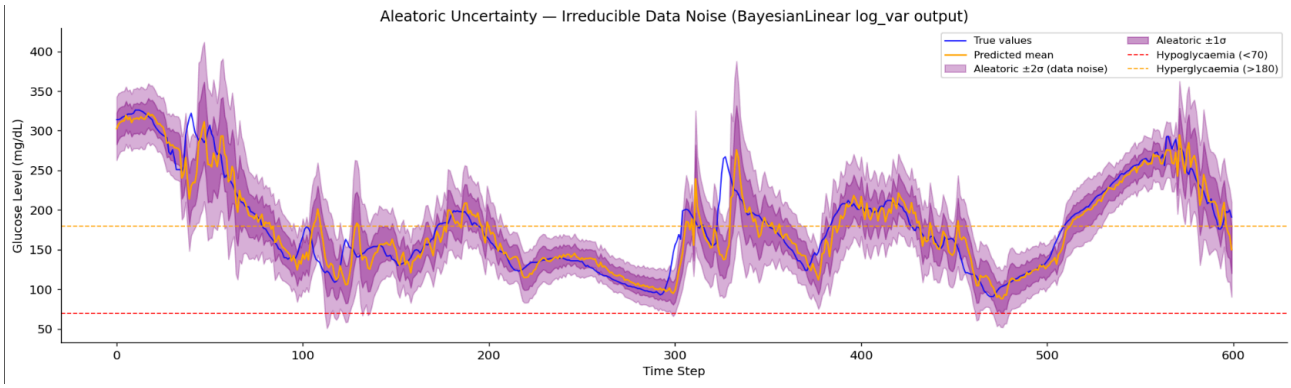
The architecture utilizes Monte Carlo (MC) Dropout to construct an empirical posterior distribution and quantify uncertainty. Unlike deterministic models, the dropout layers remain active during testing, and each input sequence is passed through the network 50 times to simulate a Bayesian stochastic process. The final predicted glucose value is computed as the posterior mean of these 50 forward passes. The variance across these passes represents the Epistemic uncertainty (model ignorance due to lack of training data), while the expectation of the Bayesian output layer's variance represents the Aleatoric uncertainty (irreducible sensor noise).

Predicted vs True Blood Glucose:



The predicted glucose values closely follow the overall trend of the true glucose levels across the test set. The model captures major rises and drops effectively, though minor deviations are present at some time steps.

Aleatoric Uncertainty



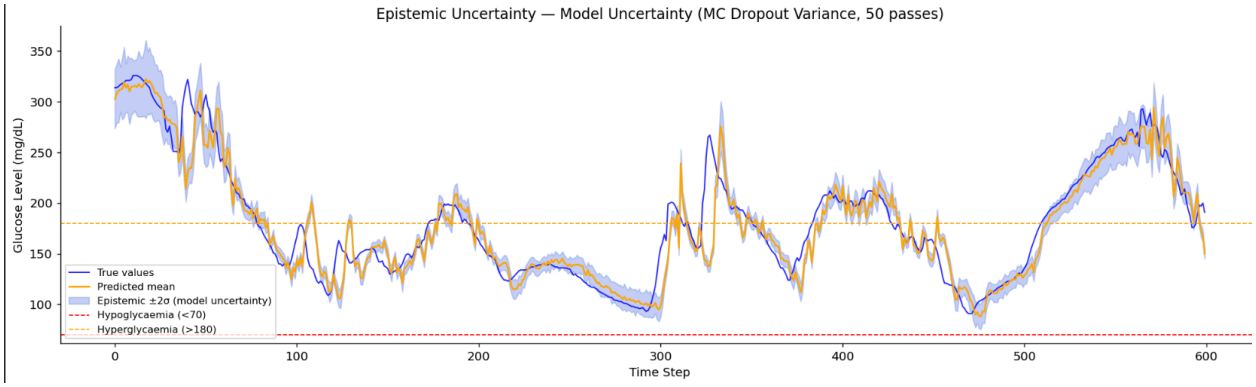
Aleatoric uncertainty represents the inherent noise in the data (e.g., sensor noise and biological variability) and cannot be reduced with more data.

The Bayesian output layer predicts both:

- Mean: expected glucose value
- Variance: uncertainty in prediction (shown by the width of the purple band)

Higher variance indicates regions where the data is more noisy or unpredictable.

Epistemic Uncertainty



Epistemic uncertainty represents the model's lack of knowledge and can be reduced with more data or better training.

It is estimated using multiple forward passes (MC Dropout), where variations in predictions indicate uncertainty in the model itself. Higher variability shows regions where the model is less confident

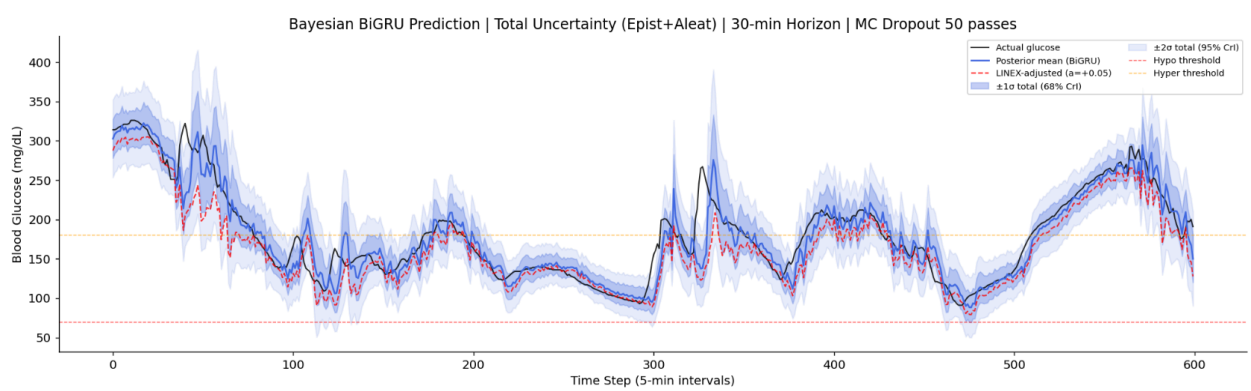
Total Uncertainty with LINEX-Adjusted Prediction

The total uncertainty combines epistemic and aleatoric components:

$$\sigma_total = \sqrt{(\sigma^2_epistemic + \sigma^2_aleatoric)}$$

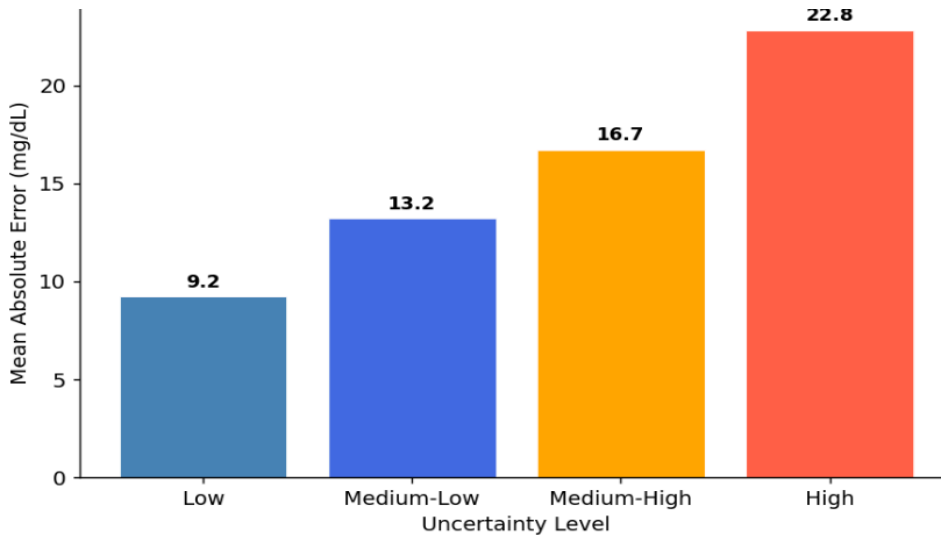
The $\pm 1\sigma$ band represents ~68% credible interval (CrI), while the $\pm 2\sigma$ band represents ~95% CrI. Empirically, the $\pm 2\sigma$ band captures the true glucose values about 96.49% of the time, closely matching the theoretical expectation.

The red dashed line shows the LINEX-adjusted prediction (a = +0.05), incorporating asymmetric risk into the model output.



Calibration - Error vs Uncertainty Analysis

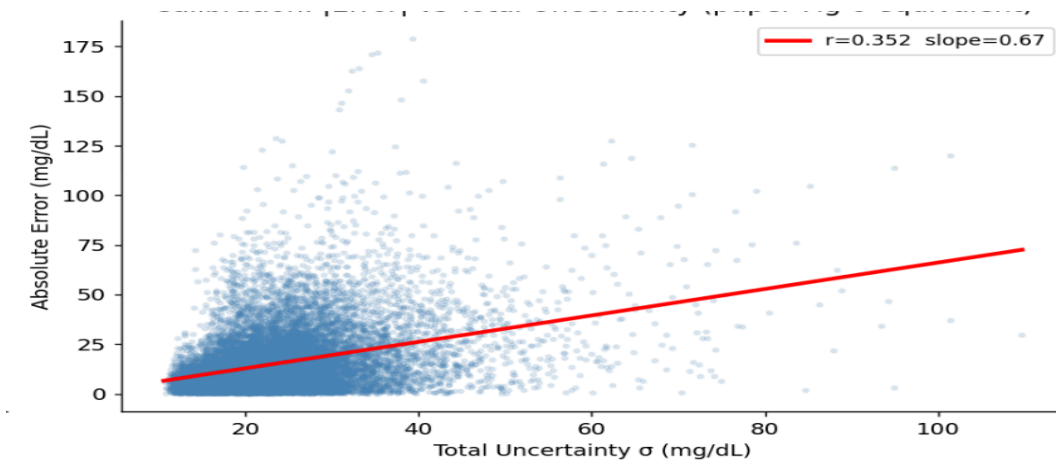
The mean absolute error increases as predicted uncertainty increases, indicating good calibration.Low-uncertainty predictions have smaller errors, while high-uncertainty predictions show larger errors, demonstrating that the model's uncertainty estimates align well with actual prediction errors.



Error vs Uncertainty Correlation

The scatter plot shows a positive correlation (r = 0.352) between predicted uncertainty and absolute error.

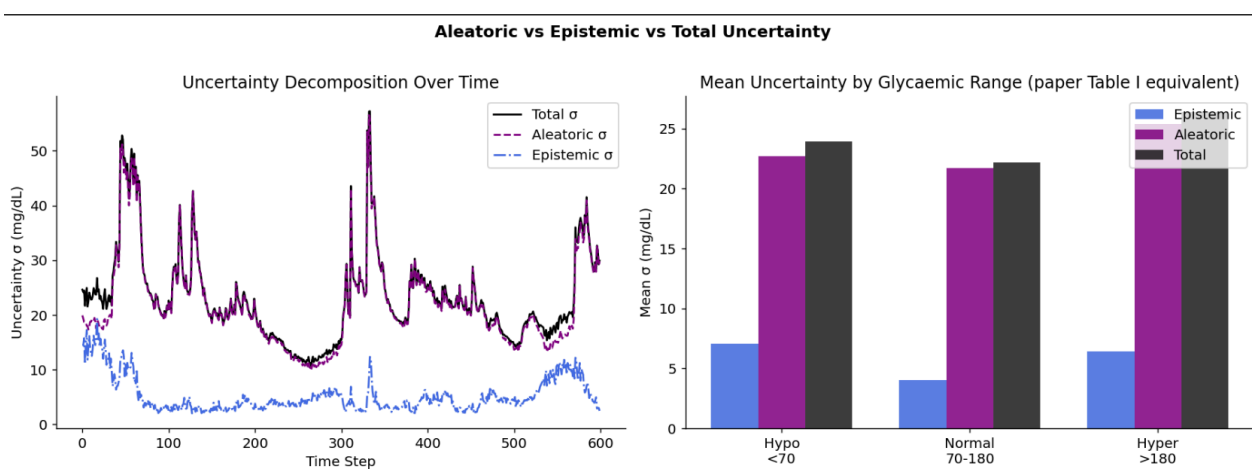
As uncertainty increases, the error generally increases, indicating that the model assigns higher uncertainty to less accurate predictions. The upward trend (slope = 0.67) supports that uncertainty estimates are meaningfully aligned with prediction errors.



Uncertainty Decomposition

The total uncertainty is composed of aleatoric and epistemic components, with aleatoric uncertainty dominating across most time steps.

Across glycaemic ranges (hypo, normal, hyper), aleatoric uncertainty remains the major contributor, while epistemic uncertainty is smaller but varies slightly, indicating the model is generally confident but data noise is the primary source of uncertainty.



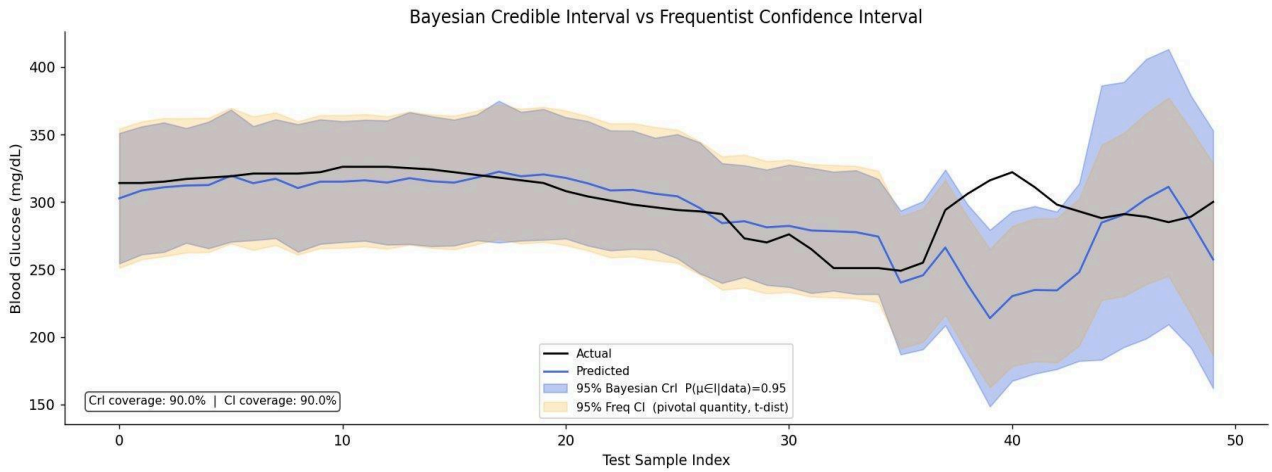
Uncertainty Across Glycaemic Ranges

Metric	Hypo (<70)	Normal (70–180)	Hyper (>180)
% of samples	2.3%	55.9%	41.8%
Mean Epistemic σ (mg/dL)	7.08	4.02	6.44
Mean Aleatoric σ (mg/dL)	22.73	21.73	25.39
RMSE by zone (mg/dL)	22.87	19.21	25.29

Uncertainty decomposition and RMSE by glycaemic zone. Overall safety margin (95% CrI coverage): 96.49%.

- Most data is normal (55.9%); hypo is very small.
- Aleatoric > Epistemic → data noise dominates.
- Error is lowest in normal, higher in extreme (hyper).
- ~96.49% coverage → good calibration.

Bayesian Credible Interval vs Frequentist Confidence Interval



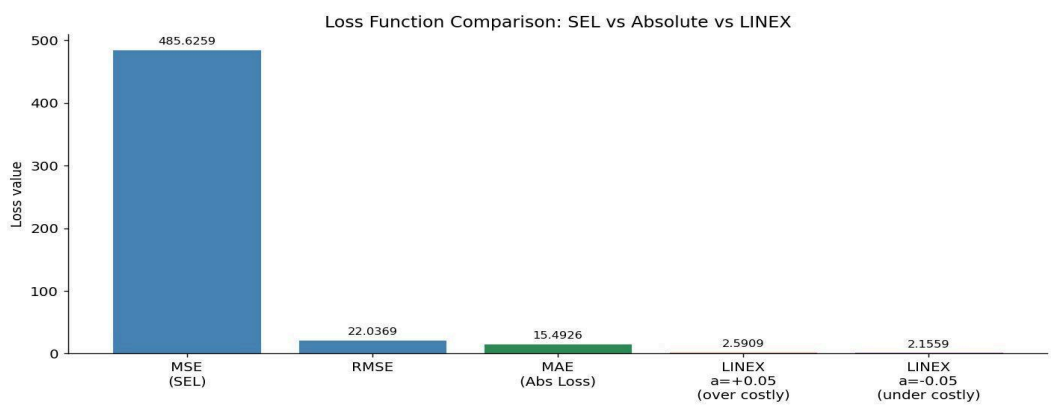
Comparison of the 95% Bayesian Credible Interval (CrI, blue) and the 95% Frequentist Confidence Interval (CI, orange) over 50 test samples.

- CrI (blue) adapts to each prediction → wider with high uncertainty, narrower when confident.
- CI (orange) uses pooled standard error → less responsive to individual predictions.
- Both achieve ~90% empirical coverage on this sample window.

Key difference:

- CrI:P(true glucose ∈ I | data) = 0.95
- CI: long-run frequency guarantee of the method

Loss Function Comparison:



Comparison of MSE (SEL), RMSE, MAE, and LINEX loss values on the test set.

The model error is evaluated using different loss functions:

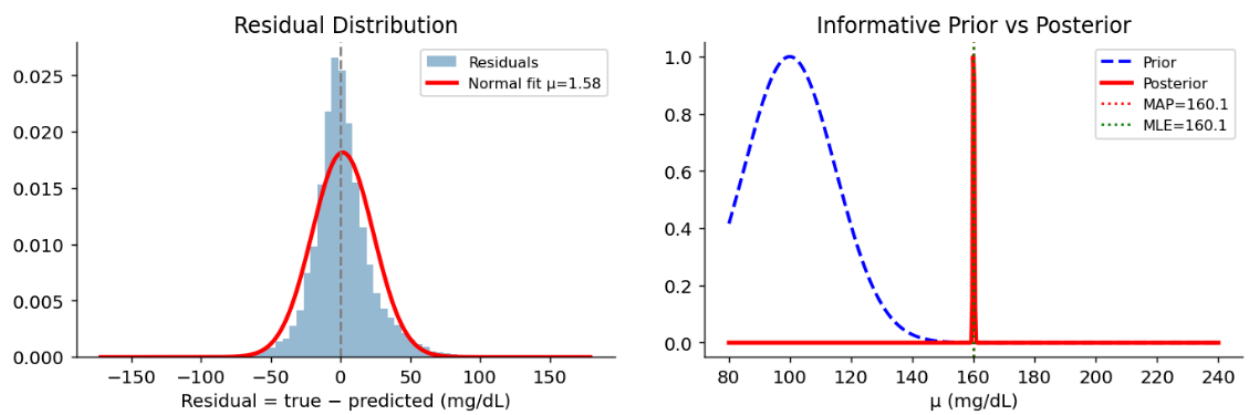
- *MSE = 485.63, RMSE = 22.04 mg/dL → symmetric squared error (RMSE is interpretable in glucose units)*
- *MAE = 15.49 mg/dL → average absolute error, robust to outliers*

- *LINEX (a = +0.05) = 2.59 → penalizes overestimation more*
- *LINEX (a = −0.05) = 2.16 → penalizes underestimation more*

The lower loss for (a = -0.05) indicates that underestimation errors are less frequent or less severe, while overestimation errors contribute more to the total loss.

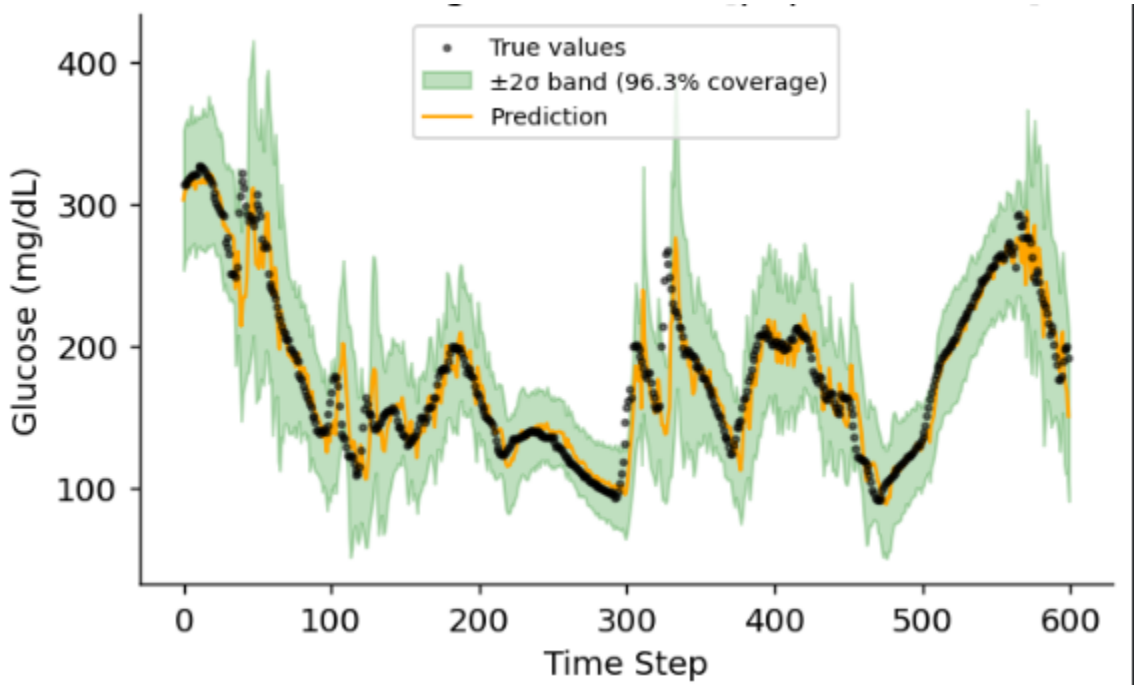
This asymmetric evaluation is important for clinical risk management, where different errors have different consequences.

Residuals & Posterior Analysis



Residuals are centered near 0 → low bias in predictions
Distribution is approximately normal → errors are well-behaved
Posterior vs Prior: posterior is sharply peaked → strong data influence
Posterior aligns with MAP ≈ MLE (~160 mg/dL) → consistent estimation

Prediction with Uncertainty Band



Predicted values closely follow true glucose trends
The ±2σ band covers ~96.3% of true values → good calibration
Wider bands indicate higher uncertainty in difficult regions

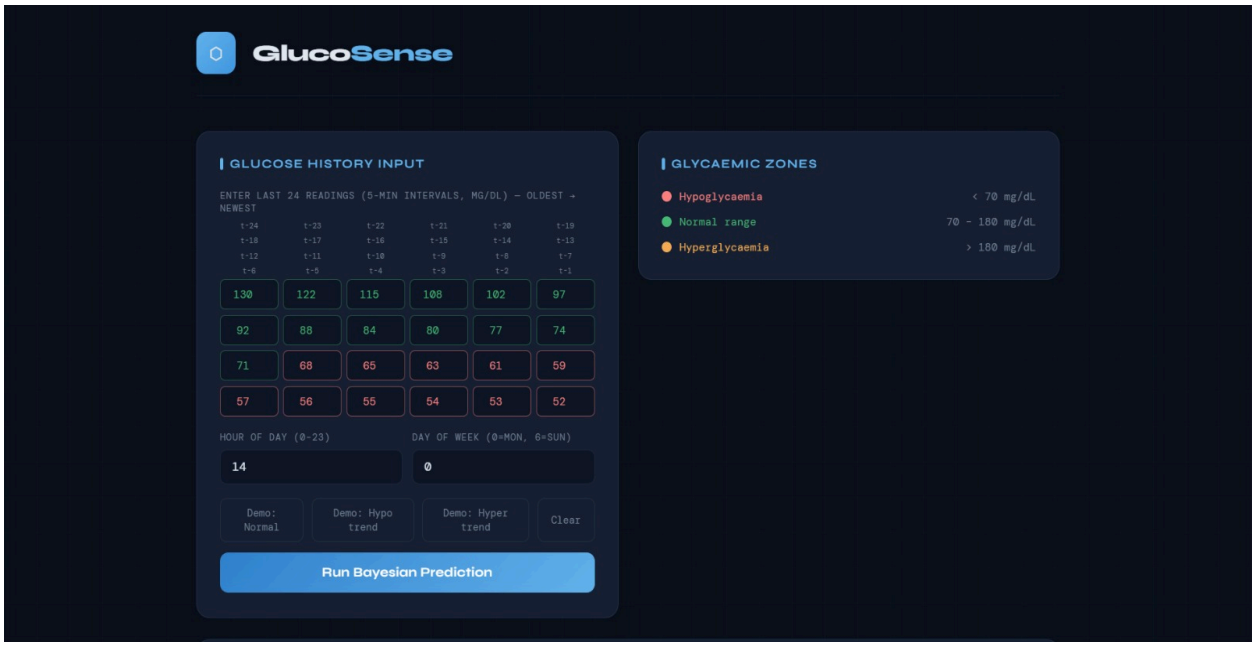
Summary of Key Metrics

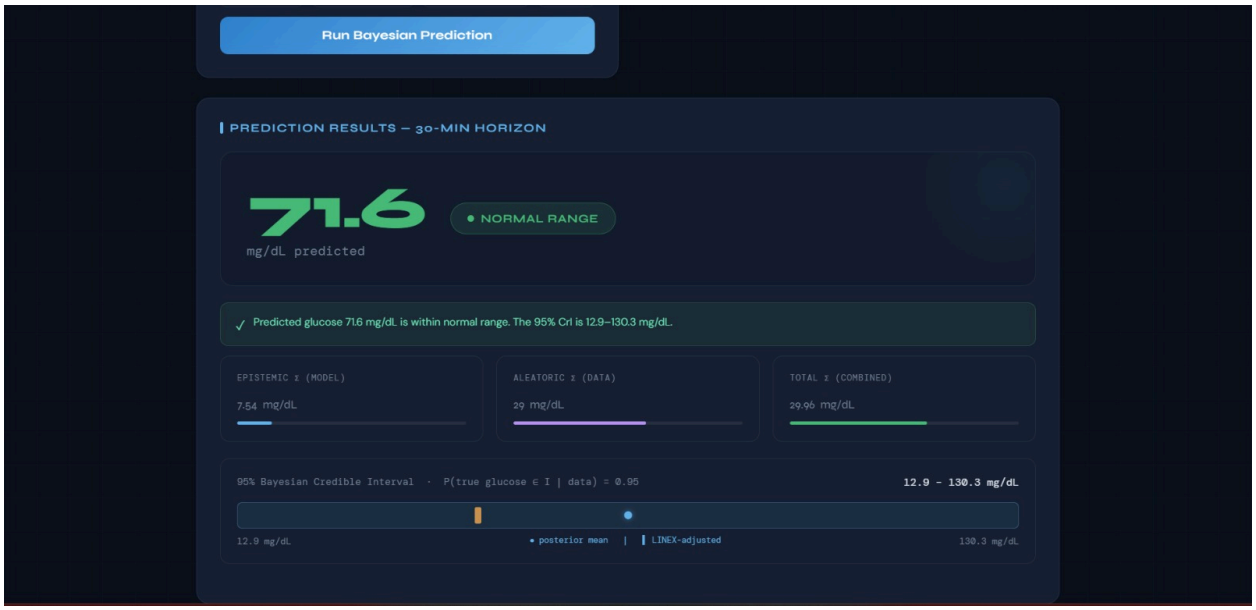
Metric	Value
RMSE (Squared Error Loss)	22.04 mg/dL
MAE (Absolute Error Loss)	15.49 mg/dL
R^2	0.9411
LINEX (a=+0.05, over penalty)	2.5909

Metric	Value
LINEX (a=−0.05, under penalty)	2.1559
±2σ Coverage	96.49%
Error–Uncertainty correlation r	0.352
Mean Epistemic σ	5.2 mg/dL
Mean Aleatoric σ	22.1 mg/dL
Best training epoch	42 (of 50 max)

Web Application — GlucoSense

A full-stack interactive web application was developed to demonstrate the model in a user-facing interface. The application is named GlucoSense and allows a user to enter 24 historical glucose readings and receive a complete Bayesian prediction with uncertainty decomposition.





Architecture

- Backend: Flask (Python) — loads the trained model and scalers, reconstructs the 14-feature input from the raw glucose history, runs 50 MC Dropout passes, and returns the full prediction as a JSON response.

- Frontend: HTML/CSS/JavaScript — a clinical dark-theme interface with real-time glucose coloring (red for hypo, amber for hyper, green for normal), animated prediction results, an interactive 95% Credible Interval bar, and a histogram of all 50 MC Dropout sample outputs.

Outputs Displayed

- Predicted glucose in 30 minutes (posterior mean, mg/dL)
- LINEX-adjusted prediction (a = +0.05)
- Epistemic σ, Aleatoric σ, Total σ with proportional bar charts
- 95% Bayesian Credible Interval shown as an interactive range bar
- MC Dropout sample histogram (50 bins) showing the full posterior distribution
- Glycaemic zone classification with clinical alert (Hypo / Normal / Hyper)
- Demo buttons for Normal / Hypo trend / Hyper trend scenarios

Conclusion

In this project, an uncertainty-aware blood glucose prediction system was developed using a **Bayesian Neural Network (BNN)** based on a Bidirectional GRU with attention. The model is able to predict future glucose levels while also quantifying the **uncertainty** associated with each prediction, making it more reliable than traditional point-estimate models.

The model achieves strong predictive performance, with RMSE ≈ 22 mg/dL and $R^2 \approx 0.94$, and closely follows the true glucose trends over time. More importantly, it provides well-calibrated uncertainty estimates, with approximately 96% of true glucose values falling within the predicted **95% Credible Interval (CrI)**, demonstrating strong **calibration**.

The uncertainty analysis shows that **Aleatoric Uncertainty** (data noise and biological variability) is the dominant component, while **Epistemic Uncertainty** (model uncertainty) remains relatively low. This indicates that the model has effectively learned the underlying patterns and that most uncertainty comes from inherent data variability.

The model is trained using a combination of **Negative Log-Likelihood (NLL)** and **KL Divergence**, which ensures both accurate predictions and proper **regularization** of the Bayesian weight distributions.

The use of **LINEX Loss (asymmetric loss function)** introduces a **decision-theoretic** perspective by penalizing overestimation and underestimation differently. This is particularly important in clinical settings where different types of errors carry different risks.

Furthermore, the comparison between **Bayesian Credible Intervals** and **Frequentist Confidence Intervals** highlights the advantage of probabilistic modeling. The **CrI adapts to each**

prediction, providing more informative and context-specific uncertainty estimates.

Overall, this work demonstrates that combining **deep learning** with **Bayesian inference** results in a model that is not only accurate but also **interpretable, calibrated, and risk-aware**. The integration of this system into the GlucoSense web application further highlights its potential for real-time, practical deployment in diabetes management.

Google Colab link: **Bayesian Glucose Prediction.ipynb**

Github:
https://github.com/eepsitavaishnavi/Blood-Glucose-Prediction-Using-Bayesian-Neural-Network

References:

1.OhioT1DM Dataset (Kaggle): OhioT1DM Dataset

2. Bayesian Neural Network for Uncertainty-Aware Blood Glucose Prediction for Type 1 Diabetes.
https://share.google/wBmJRvxiuvFX2CNwg

3. Y. Gal and Z. Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning" ICML, 2016.